Assignment-6

Object Oriented Programming Lab

(Java: Inheritance)

1. You are required to write a base class Student having details (name, registration number, father's name, address (current address, permanent address), and contact details (phone number, email Id). UG Student, PG Student, and PhD Student extend the Student class they have details about the past qualifying degree and marks. Make each attribute protected and provide suitable getter and setter to access attributes.

2. Write another base class Course which stores details about the courses registered by a Student. Course has attributes such as code, name, and credit. UG Course and PG Course extends the Course class. The course code follows the following syntax PG-xxxx and UG-xxxx here x is a number which uniquely identifies a course. Make each attribute protected and provide suitable getter and setter to access attributes.

3. Write another base class Payment which stores the payment details of the registered Student. Payment has a single attribute payment ID which is unique for each payment. Payment DD, Payment UPI, and Payment NB extends the Payment class, these classes' stores details of payment using Demand Draft, UPI, and Net Banking. Add relevant attributes to these classes to store payment details. Make each attribute protected and provide suitable getter and setter to access attributes.

4. Add one instance of each Course class and Payment class to the student class so that course and payment details of a student can be associated with each student. Add a member function in Student class to print course detail and payment details using student's registration number. Demonstrate the program by creating 10 students and printing their details.

5. Write a class Employee having attributes: employee ID, name, department, dob, designation, year Of Joining, and phone Number. Add a function to print details of employee using employee ID. Two classes, i.e., Faculty and Office-Staff, extends the Employee class. The Faculty has a list of subjects and labs taught by them Office-Staff has a list of skills which stores skill such as Typing, Technician, etc. Each of the above classes has appropriate getter setter for inserting skill in the list and printing the list of skill. Demonstrate above class using a Test class which creates 5 employees of di erent types and print their details.

6. Classes HOD, DUGC, DMPC, and DDPC extends the Faculty class (given in the above question). These classes stores detail specific to the Head of the Department (HOD), Department Undergraduate Committee (DUGC) Convenor, Department Master Program Committee (DUGC) Convenor, and Department Doctoral Program Committee (DUGC) Convenor. Skilled Staff and Unskilled Staff extends class Office-Staff (given in the above question). Each class has a public method to String() which returns their details as a String. Demonstrate above class using a Test class which creates 6 employees of different types and print their details.

7. Employee and **Production Worker** Classes

Design a class named Employee. The class should keep the following information in fields:

- Employee name,
- Employee number in the format XXX {L, where each X is a digit within the range 0-9 and the L is a letter within the range A-M.
- Hire date

Write one or more constructors and the appropriate accessor and mutator methods for the class. Next, write a class named Production Worker that extends the Employee class. The Production Worker class should have fields to hold the following information:

Shift (an integer)

Hourly pay rate (a double)

The workday is divided into two shifts: day and night. The shift field will be an integer value representing the shift that the employee works. The day shift is shift 1 and the night shift is shift 2. Write one or more constructors and the appropriate accessor and mutator methods for the class. Demonstrate the classes by writing a program that uses a **Production Worker** object.

8. **Shift Supervisor** Class

In a particular factory, a shift supervisor is a salaried employee who supervises a shift. In addition to a salary, the shift supervisor earns a yearly bonus when his or her shift meets production goals. Design a Shift Supervisor class that extends the Employee class you created in question 7. The Shift Supervisor class should have a field that holds the annual salary and a field that holds the annual production bonus that a shift supervisor has earned. Write one or more constructors and the appropriate accessor and mutator methods for the class. Demonstrate the class by writing a program that uses a Shift Supervisor object.

9. **Team Leader** Class

In a particular factory, a team leader is an hourly paid production worker that leads a small team. In addition to hourly pay, team leaders earn a xed monthly bonus. Team leaders are required to attend a minimum number of hours of training per year. Design a Team Leader class that extends the Production Worker class you designed in question 7. The Team Leader class should have fields for the monthly bonus amount, the required number of training hours, and the number of training hours that the team leader has attended. Write one or more constructors and the appropriate accessor and mutator methods for the class. Demonstrate the class by writing a program that uses a Team Leader object,